

Chapter 02 – Values, Types, Variables, and Operators

Full Exam Paper with Answers

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Part I – Multiple Choice Questions (20 Marks)

1. Which of the following is a primitive data type in JavaScript?

- A) Array
- B) Function
- C) String
- D) Date

■ Correct answer: C

2. Non-primitive data types are also called:

- A) Value types
- B) Reference types
- C) Basic types
- D) Static types

■ Correct answer: B

3. Which keyword declares a constant variable?

- A) var
- B) let
- C) const
- D) define

■ Correct answer: C

4. What is the output of `typeof 42`?

- A) string
- B) number
- C) integer
- D) float

■ Correct answer: B

5. A variable declared but not assigned a value has the type:

- A) null
- B) undefined
- C) empty
- D) void

■ Correct answer: B

6. What does the modulus operator (%) do?

- A) Divides two numbers
- B) Gives remainder after division
- C) Finds power
- D) Joins numbers

■ Correct answer: B

7. Which symbol is used for concatenating strings?

- A) -
- B) *
- C) +
- D) &

■ Correct answer: C

8. What is the result of `5 == '5'` ?

- A) true
- B) false
- C) undefined
- D) error

■ Correct answer: A

9. What is the result of `5 === '5'` ?

- A) true
- B) false
- C) undefined
- D) error

■ Correct answer: B

10. The logical AND operator is represented as:

- A) ||
- B) &&
- C) !
- D) ==

■ Correct answer: B

11. The logical OR operator is represented as:

- A) ||
- B) &&
- C) !
- D) ==

■ Correct answer: A

12. The NOT operator reverses a boolean value and is represented as:

- A) &
- B) ||
- C) !
- D) ==

■ Correct answer: C

13. Which operator increases a variable's value by one?

- A) ++
- B) +=
- C) --
- D) **

■ Correct answer: A

14. The conditional (ternary) operator is written as:

- A) if/else
- B) ? :
- C) switch
- D) ==

■ Correct answer: B

15. In JavaScript, `typeof null` returns:

- A) null
- B) object
- C) undefined
- D) string

■ Correct answer: B

16. Which keyword was used in older JavaScript but less recommended today?

- A) const
- B) let
- C) var

D) static

■ Correct answer: C

17. The result of $15 \% 2$ is:

A) 0

B) 1

C) 2

D) 7

■ Correct answer: B

18. The result of `typeof [1,2,3]` is:

A) array

B) object

C) list

D) undefined

■ Correct answer: B

19. Implicit type conversion in `'10' * 2` gives:

A) '102'

B) 12

C) 20

D) NaN

■ Correct answer: C

20. Which of the following increases value after usage (post-increment)?

A) `++x`

B) `x++`

C) `--x`

D) `x--`

■ Correct answer: B

Part II – True or False (20 Marks)

1. Data types define the kind of value stored in a program.

Answer: True

2. Primitive types include Array and Object.

Answer: False

3. String and Boolean are primitive data types.

Answer: True

4. The `typeof` operator returns the data type of a value.

Answer: True

5. Undefined means a variable has no assigned value.

Answer: True

6. The modulus operator gives the quotient of division.

Answer: False

7. The `+` sign can be used to join strings.

Answer: True

8. `==` checks both value and type.

Answer: False

9. === checks both value and type equality.

Answer: True

10. The logical AND operator is written as ||.

Answer: False

11. The OR operator returns true if one condition is true.

Answer: True

12. The NOT operator (!) changes true to false and vice versa.

Answer: True

13. ++x increases value before use.

Answer: True

14. x++ decreases the value of a variable.

Answer: False

15. The ternary operator uses ? and : symbols.

Answer: True

16. typeof null returns 'null'.

Answer: False

17. Arrays are considered objects in JavaScript.

Answer: True

18. Implicit conversion means automatic type conversion.

Answer: True

19. Explicit conversion means manual type conversion.

Answer: True

20. JavaScript variables are case-sensitive.

Answer: True

Part III – Direct Questions (20 Marks)

1. Define data type.

Answer: It defines the kind or category of a value stored or used in a program.

2. List two categories of JavaScript data types.

Answer: Primitive and Non-Primitive (Reference types).

3. Give three examples of primitive data types.

Answer: String, Number, Boolean.

4. Give two examples of non-primitive data types.

Answer: Array and Function.

5. What keyword is used to declare a constant?

Answer: const.

6. Write two differences between let and var.

Answer: let is block-scoped; var is function-scoped and older.

7. What are valid characters to start a variable name?

Answer: Letter, underscore (`_`), or dollar sign (`$`).

8. What does the `typeof` operator do?

Answer: It returns the data type of a value or variable.

9. What is the use of the modulus operator?

Answer: It gives the remainder after division.

10. What does `'=='` compare?

Answer: Only values, not data types.

11. What does `'==='` compare?

Answer: Both value and type.

12. Write the symbol for logical AND.

Answer: `&&`.

13. Write the symbol for logical OR.

Answer: `||`.

14. What does the NOT operator do?

Answer: It inverts the truth value of an expression.

15. Write an example using the ternary operator.

Answer: `let msg = (x > 5) ? 'Yes' : 'No';`

16. Differentiate between pre-increment and post-increment.

Answer: Pre-increment (`++x`) increases before use; post-increment (`x++`) after use.

17. What does `typeof null` return?

Answer: `object` (special case).

18. Define implicit type conversion.

Answer: Automatic type change by JavaScript during execution.

19. Define explicit type conversion.

Answer: Manual type change by developer using functions like `Number()` or `String()`.

20. Why are variable names case-sensitive?

Answer: Because JavaScript treats uppercase and lowercase letters as different identifiers.